# CLOUD COMPUTING FUNDAMENTALS REPORT

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## Introduction

- Distributed and cloud computing systems are built over a large number of autonomous computer nodes. These nodes machines are interconnected by SANs, LANs, or WANS in a hierarchical manner. With today's networking technology, a few LAN switches can easily connect hundreds of machines as a working cluster.

## Cloud Service Models

There are the following three types of cloud service models

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)

### 1. Infrastructure as a Service (IaaS)

- Infrastructure as a Service is a service model that delivers computer infrastructure on an outsourced basis to support various operations.
- It is also known as Hardware as a Service. IaaS customers pay on a per-user basis, typically by the hour, week, or moth. Some providers also charge customers based on the amount of virtual machine space they use.
- It simply provides the underlying operating systems, security, networking, and servers for developing such applications, and services, and deploying development tools, databases, etc.
- Examples: Digital Ocean, Amazon Web Services.

### Advantages

- Cost-Effective
- Website hosting
- Security
- Maintenance

### Disadvantages

- Limited control over infrastructure
- Security concerns

- Limited access

2. **Platform as a Service (PaaS)**
   - PaaS is a category of cloud computing that provides a platform and environment to allow developers too build applications and services over the internet.
   - PaaS servicers are hosted in the cloud and accessed by users simply via their web browser.
   - Examples: Google App Engine, Heroku

### Advantages

- Simple and convenient for users
- Cost-Effective
- Efficiently managing the lifecycle
- Efficiency

### Disadvantages

- Limited control over infrastructure
- Dependence on the provider
- Limited flexibility

3. **Software as a Service (SaaS)**
   - Software as a Services is a way of delivering services and applications over the Internet Instead of installing and maintain software, we simply access it via the Internet, freeing ourselves from the complex software and hardware management.
   - It provides a complete software solution that you purchase on a pay-as-pay-go basis from cloud service provider.
   - Examples: Gmail, Google Drive, Zoom, Netflix

### Advantages

- Cost-Effective
- Reduced time
- Accessibility
- Automatic updates
- Scalability

### Disadvantages

- Limited customization
- Security concerns
- Limited control over data

## Cloud Provider

- A cloud provider is responsible for making a service available to the cloud consumer.  Cloud provider may be a person, team or an organization.

## Major Providers

1. AWS - Amazon Web Services

2. Microsoft Azure

3. Google Cloud Platform – GCP

4. IBM Cloud

5. Oracle Cloud

## Comparisons of Two Cloud Providers

| Feature | Amazon AWS | Microsoft Azure |
|---|---|---|
| Founded | 2006 | 2010 |
| Market Leader | Yes | Yes |
| Free Tier | 12 months free | 12 months free + $200 credit |
| Popular Services | EC2, S3, Lambda | Virtual Machine, Blob Storage |
| Best For | Startups, Largescale apps | Enterprises, Window apps |

# CONCLUSION